Making the remarkable regular: 'Marked absolutive' in Nias (to appear in *Glossa*)

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Abstract This paper re-examines the unusual 'marked absolutive' argument marking pattern in Nias Selatan. While previous works propose that Nias instantiates a 'marked absolutive' case pattern (Brown 2001; 2005; Baker 2015), this paper argues against this approach. Instead, Nias' argument marking system is shown to be a morpho-phonological process, which is unrelated to case, and that Nias' 'marked absolutive' pattern is due to a conspiracy of this process' sensitivity to syntactic domains (i.e. phases) interacting with independently active syntactic operations unrelated to case. Moreover, it is shown that this analysis is consequential in regards to the typology of attested case patterns. In addition, it is shown that Nias' argument marking system is an instance of a class of cross-linguistically attested morpho-phonological processes that are sensitive to syntactic domains in the same manner.

Keywords: syntax-phonology interface; phases; case typology; PF domains

The empirical focus of this paper concerns the unusual argument marking patterns observed in Nias Selatan (Austronesian; Indonesia) and its typological and theoretical implications for theorizing about case patterns. As illustrated in (1) and (2), Nias core arguments participate in an argument marking pattern (glossed as MUT), where the morpho-phonological form of the nominal undergoes a change that is seemingly dependent on grammatical role. Here, intransitive subjects and objects are marked to the exclusion of transitive subjects, which systematically surface unmarked.[1]

(1) ma=aso?a **d**uhituhi
PERF=fall MUT.trivet
'The trivet stone fell.' [B559]
***t**uhituhi* → ***d**uhituhi*

(2) ?i-a **ʙ**avi ?ama Gumi
3SG.RLS-eat MUT.pig father Gumi
'Father Gumi eats pig.' [DB61]
***b**avi* → ***ʙ**avi*

Due to this seeming connection between the surface form of the argument and its grammatical role, several works have noted that Nias' argument marking pattern resembles case marking patterns typically associated with ergative/absolutive case systems (Donohue & Brown 1999; Brown 2001; 2005; Baker 2015). That is, languages where the morphological form of intransitive subjects and objects (i.e. absolutive arguments) are grouped distinct from the morphological form of transitive subjects (i.e. ergative arguments). Building on this observation, such works have proposed that the argument marking pattern in (1) and (2) is, in fact, the realization of a 'marked absolutive' case system. Specifically, that the morpho-phonological change observed on intransitive subjects and objects is the realization of absolutive case (ABS) and that ergative case (ERG) is unmarked.

[1] All Nias examples in this paper are from Brown 2001 and Donohue & Brown 1999. Beside Nias examples, the citation convention I use is B or DB, followed by page number, e.g. [B559] in example (1). In all Nias examples, I follow the orthography in Brown 2001, except in some cases where I diverge to more closely reflect phonological form, e.g. in Brown 2001 <kh> in orthography represents the phoneme [x], and <ndr> for [dʳ], whereas in this paper I use the phoneme. For discussion of Nias orthographic conventions and its relation to phonetic description see Brown 2001: 21-49. Glosses follow Leipzig glossing conventions. Additional glossing abbreviations follow Brown 2001; see Abbreviations.

While a case-alignment based approach to Nias' argument marking pattern appears to account for the apparent sensitivity to grammatical role in (1) and (2), I argue that this is not the correct approach. First, from a typological standpoint, such case-based approaches render Nias' argument marking pattern an extreme typological outlier. In typological works on absolutive case patterns, it has been generally observed that either ERG is the marked case and ABS is unmarked or both ERG and ABS are marked with distinct forms. Crucially, across attested absolutive patterns, no language has been observed to exhibit a 'marked absolutive' pattern, where ABS is marked and ERG is the unmarked case (Dixon 1994; Deal 2014). Thus, on this approach, Nias' 'marked absolutive' pattern (or unmarked ergative), would be the only attested instance of such a pattern. Given this situation, I propose that this is good reason to reevaluate Nias' argument marking system as the realization of absolutive case. Second, from a language-internal standpoint, I show that a case-based approach to Nias' argument marking pattern fails to predict the distribution of marked nominals outside of the basic case in (1,2). Specifically, that while the morpho-phonological marking may superficially appear to correlate with absolutive grammatical role in cases like (1,2), this correlation systematically breaks down across a range of cases. More generally, I show that the distribution of marked arguments in Nias does not clearly correspond to the prototypical distribution of ABS-marked arguments observed cross-linguistically. Further, I show that the distribution of marked arguments in Nias does not, in general, correspond to any type of grammatical role.

Given the above issues, I will propose a reevaluation of Nias' 'marked absolutive' case system, showing that what conditions argument marking is not the grammatical role of the argument but, rather, the structural position the nominal is located in at spell-out to PF. Specifically, that argument marking is a morpho-phonological process that occurs when the nominal and trigger are located in the same phase domain, and that the apparent 'marked absolutive' pattern is simply a PF reflex of transitive subjects systematically being outside of the argument-marking domain. On this analysis, Nias' argument marking system is situated within the class of morpho-phonological processes that are sensitive to syntactic domains, e.g. external sandhi processes (Seidl 2001; d'Alessandro & Scheer 2015; Bošković 2017, *i.a.*). More specifically, the morpho-phonological process that underlies Nias' mutation pattern is on a par with more familiar mutation processes, e.g. mutation in Welsh, which display the same kind of sensitivity to syntactic contexts regarding where such morpho-phonological processes may occur (Pyatt 1997; 2003; Roberts 1997; Tallerman 2006; Borsley et al. 2007, *i.a.*).

To summarize, the first overarching goal of the paper is to show that Nias' argument marking system is not the morphological realization of a 'marked absolutive' case system. Rather, it is the result of a morpho-phonological process that is unrelated to case. The second goal is to develop an analysis of Nias' argument marking pattern in these morpho-phonological terms. That is, to develop an analysis where argument marking is the result of a domain-based morpho-phonological process that is regulated by the syntactic position of the nominal and triggering element. I show that this approach, where argument marking is a morpho-phonological process of this type, has a far greater empirical coverage than the case approach. Additionally, on an approach where argument marking is a morpho-phonological surface effect, the typology of attested case patterns is further restricted by eliminating the only attested 'marked absolutive' case pattern.

The paper is structured as follows. §1 outlines the argument marking paradigm for core arguments and the case approach to the pattern and presents the empirical and theoretical challenges for the case approach. In §2, I develop the new approach to Nias argument marking and shows that this approach captures the empirical distribution of argument marking and situates Nias' system within a broader cross-linguistic pattern of mutation phenomena. §3 concludes.

1 Argument marking patterns and the case analysis

As (3) illustrates, default word order in Nias is VO(X)S, where internal arguments surface clause-medially and subjects consistently surface clause-finally. Additionally, as in (4), Nias productively allows for null subjects (Brown 2001; 2005).

(3) ʔi-beʔe gefe xö zodʳöröu ʔama Dali
3SG.RLS-give MUT.money DAT MUT.healer Ama Dali
'Ama Dali gave money to the village healer.' [B269]

(4) ʔu-sura zura *e*
1SG-write MUT.letter
'I wrote a letter.' [B319]

Donohue & Brown (1999) and Brown (2001; 2005) observe that core arguments in Nias participate in an argument marking pattern— termed 'nominal mutation' (Brown 2001; 2005)— where nominals undergo a morpho-phonological change to the initial segment of the stem, as in Tab. 1.[2]

Unmarked	V	f	t	s	c[ʧ]	k	ʔ	b	d	m	n	l	r	dʳ
Marked	n-V	v	d	z[ʤ]	z[ʤ]	g	g	ʙ	dʳ	-	-	-	-	-

Table 1: Mutation Pattern.

In the case of intransitive clauses, the subject is consistently marked. As (5a) illustrate, the unergative subject *iβania* ('her sister') surfaces with a prothetic nasal, i.e. *niβania*. Similarly, in (5b), the initial [t] of unaccusative subject *tuhituhi* ('trivet stone') undergoes voicing, i.e. *duhituhi*.

(5) a. m-eʔe **n**iβa-nia
DYN-cry MUT.sibling-3SG.POSS
'Her sister is crying.' [B118]

b. ma=asoʔa **d**uhituhi
PERF=fall MUT.trivet
'The trivet stone fell.' [B559]

In the case of transitive clauses, objects are marked in the same way as intransitive subjects. In (6a), the object *sawi* undergoes voicing surfacing as *zawi* and in (6b) the object *bavi* undergoes trilling surfacing as *ʙavi*. Importantly, in the case of transitive subjects, these arguments consistently surface unmarked, i.e. *Fasui* and *ʔama Gumi* in (6a) and (6b).

(6) a. ma=i-uri **z**awi Fasui.
PERF=3SG.RLS-keep MUT.cattle Fasui
'Fasui kept cattle.' [B366]

b. ʔi-a **ʙ**avi ʔama Gumi
3SG.RLS-eat MUT.pig father Gumi
'Father Gumi eats pig.' [DB61]

What the data appear to show is that the surface form of core arguments in Nias appear to be conditioned by grammatical role. In the case of absolutive arguments (i.e. intransitive subjects and objects) the argument is marked; in the case of ergative arguments (i.e. transitive subjects) the argument surfaces unmarked.

[2] As in Tab. 1, when orthography and IPA conventions diverge in tables, I include the IPA representation in parentheses beside the orthography.

1.1 The case analysis

As was shown, in Nias intransitive subjects and objects, are grouped distinctly from transitive subjects with respect to morphological form. In this respect, Nias' argument marking pattern seemingly parallels ergative-absolutive case patterns attested cross-linguistically. That is, languages where case morphology distinguishes ergative arguments from absolutive arguments (Blake 1994; König 2008). To illustrate, consider Warlpiri where intransitive subjects (7a) and objects (7b) are unmarked, while transitive subjects are marked with the ERG suffix *-ngku* (7b).

(7) a. kurdu ka wanka-mi
child.ABS PRS.IMPF speak-NPST
'The child is crying.'

b. Ngarrka-**ngku** ka wawirri panti-rni
man-ERG PRS.IMPF kangaroo.ABS spear-NPST
'The man is spearing the kangaroo.' (Hale 1983: 13,6)

Similarly, Niuean intransitive subjects (8a) and objects (8b) are marked with the ABS particle *e*, while transitive subjects (8b) are marked with the ERG particle *he*.

(8) a. Ne fano **e** tehina haaku.
PST go ABS brother POSS
'My little brother went.'

b. Ne kai **he** puti ia **e** moa.
PST eat ERG cat DEM ABS chicken
'The cat ate the chicken.' (Seiter 1980: 29)

In both Warlpiri and Niuean absolutive arguments are morphologically distinguished from ergative arguments. In this respect, Nias' argument marking pattern parallels Warlpiri and Niuean's patterns. As shown, Nias intransitive subjects (5a,5b) and objects (6a,6b), are morphologically distinguished from transitive subjects (6a,6b), by undergoing a morpho-phonological change to the initial segment of the nominal.

Given this surface parallel between absolutive case systems and Nias' argument marking pattern, previous works have hypothesized that Nias realizes a 'marked absolutive' case system (Donohue & Brown 1999; Brown 2001; 2005; Baker 2015). Specifically, that the marking observed on absolutive arguments is the morphological realization of [ABS] and the absence of such marking with [ERG]—i.e. the morphologically unmarked case, as in (9), where MUT represents the abstract morpheme that triggers the phonological change (as shown in Tab. 1).

(9) Case exponents:
a. [ABS] ↔ MUT
b. [ERG] ↔ ∅

Building on the case hypothesis, Baker (2015) proposes that the distribution of marked arguments in Nias is due to a specific parameterization of dependent case assignment, where ABS is the dependent or 'marked' case in Nias. On this proposal, ABS is assigned to an argument in a case assignment domain in contexts where that nominal does not c-command another argument.[3]

(10) **Nias marked ABS dependent case assignment:**

[3] For the purpose of the present discussion, I am assuming that the case rule applies at Spell-out when syntactic structure is sent to PF (Baker 2015) (cf. Marantz 1991).

a. Assign NP_1 ABS, if there is no other NP_2 in the same dependent case domain (i.e. IP) as NP_1, such that, NP_1 c-commands NP_2;
b. Otherwise, NP_1 is ergative, i.e. unmarked. (Syntax)

(11) [ABS] → MUT (PF)

On this approach, the case rule in (10) assigns ABS to intransitive subjects and objects due to not c-commanding another NP inside the IP case assignment domain. Since transitive subjects c-command the object, transitive subjects are unmarked.

Importantly, on this proposal, case assignment in Nias closely parallels case assignment in other, typologically more typical, 'marked ergative' languages, e.g. Warlpiri and Niuean. In such languages, the case algorithm is also sensitive to c-command relations between arguments. However, the key difference is that while in Nias the case assignment algorithm is parameterized to mark arguments that do not c-command other arguments, i.e. absolutives, in marked ergative languages the algorithm is parameterized to mark arguments that do c-command other arguments, i.e. ergative arguments, as in (12).

(12) **Marked ERG dependent case assignment:**
a. Assign NP_1 ERG only if, there is an NP_2 in the same case domain (i.e. IP) as NP_1, such that, NP_1 c-commands NP_2;
b. Otherwise, NP_1 is absolutive.

Thus, the surface marking pattern observed with Nias' arguments is derived in terms of an underlying absolutive case system paralleling other absolutive case systems, e.g. Warlpiri and Niuean. On this analysis, the difference is that in Nias ABS is the marked case and in other languages ERG is. More generally, on the case hypothesis, the marking observed on absolutive arguments (and its absence on ergatives) is directly tied to the presence/absence of [ABS], which is realized as mutation, as in Tab. 1.

1.2 Challenges for case

While the case hypothesis has a degree of plausibility in the case of the surface pattern observed with core arguments, there are a range of syntactic contexts where an argument is unmarked despite being a prototypical absolutive argument. Given that on the case approach the absence of argument marking is the zero-realization of ERG, this is not expected.

As shown, clause-initial nominals, e.g. topicalized (13) and focused nominals (14), consistently surface unmarked. In both (13,14) the arguments *siʔo* and *jaʔugö* are both absolutive arguments, i.e. objects, but do not surface marked.

(13) siʔo höʔö ma=i-taruʔ-ö ba danö
stick DIST.DEM PERF=3SG.RLS-plant-TR LOC MUT.ground
'That stick, he planted in the ground.' [B361]

(14) ha jaʔugö ʔö-fa-manömanö-si
only 2SG 2SG.RLS-DYN-talk-APP
'You're just talking about yourself.' [B175]

If argument marking is the realization of ABS and its absence the realization of ERG, then topicalization/focalization of an argument, as in (13,14), should not result in a change in case form, i.e. a switch to ERG.

Additionally, relativized arguments also consistently surface unmarked. As shown in (15a,15b), the intransitive subjects (i.e. absolutive argument) have undergone relativization to a higher clause-initial position and surface unmarked.

(15) a. so samösa si=möi
EXIST someone REL=go
'There's someone who is coming.' [B135]

b. ʔadu s=a-ma-kobu
statue REL=IMPV-DYN-sit.on.haunches
'The statue of the man sitting on his haunches...' [B240]

Assuming that relativization in (15a,15b) is due to Ā-movement to a higher position in the clause, movement to a higher position should not result in a change in case form. Cross-linguistically, this is what is observed regarding the distribution of ABS-marking under Ā-movement. As illustrated in (16a,16b) with Niuean, ABS-marked arguments retain their case morphology under Ā-movement, e.g. relativization. In (16a,16b), the absolutive arguments have been relativized and the ABS particle *e* is retained.

(16) a. **e** mena (ne) kua taute e Sione
ABS thing PRS PERF ERG Sione
'the thing that Sione has fixed'

b. **e** tama ne hau i Makefu
ABS child PRS come from Makefu
'the child who comes from Maekfu' (Massam 1995: 87, 86)

Thus, on an approach where the presence/absence of morphological marking on nominals is the realization of case in Nias, the observation that this marking is lost under Ā-movement is highly unexpected.

Finally, promoted subjects in passive constructions, quite generally surface unmarked in Nias. As (17a,18a) show, the promoted subject surfaces in a clause-initial position. However, in both (17a,18a) the promoted subject surfaces unmarked (cf. (17b,18b)).

(17) a. ʔahe βatö ni-tefe=gu ʔidanö
floor PASS-sprinkle=1SG.POSS water
'The floor that I sprinkled with water. [B553]

b. ʔu-tefe gahe βatö ʔidanö
1SG.RLS-sprinkle MUT.floor water
'I sprinkled the floor with water.' [B552]

(18) a. baʔa ni-fönu-i=nia ʔidanö
tub PASS-fill-TR=3SG.POSS water
'The tub which she filled with water.' [B554]

b. ʔi-fönu-i βaʔa ʔidanö
3SG.RLS-fill-TR MUT.tub water
'She filled the tub with water.' [B553]

Assuming that the Nias passive involves valency reduction, i.e. detransitivization, on the case hypothesis, the promoted subjects in (17a,18a) should bear ABS and, thus, surface marked. However, as (17a,18a) show, under passivization the promoted subject surfaces unmarked. Further, given that ERG is the putative morphologically unmarked case, this would suggest that under the passive, the promoted subject is marked with ERG, which is highly unexpected given that ERG marks nominals that are transitive subjects and, thus, should be absent on the subject in (17a,18a).

In addition to the cases above, where a nominal surfaces unmarked despite being a prototypical absolutive argument, there are a range of contexts where a nominal surfaces marked but is not a prototypical absolutive argument.

The first case where nominals are marked despite not being absolutive arguments is with oblique arguments. That is, arguments which are introduced by a preposition and typically surface with a case form associated with the preposition, e.g. dative (DAT) or locative (LOC) case. As shown in (19a,19b), nominals preceded by the dative preposition *xö* must be marked. Similarly, nominals preceded by the locative preposition *ba*, as in (19c) must be marked.

(19) a. ʔI-beʔe gefe [xö **z**odʳöröu] ʔama Dali
3SG.RLS-give MUT.money DAT MUT.healer Ama Dali
'Ama Dali gave money to the village healer.' [B269]

b. La-tema ganaʔa [xö **z**iʔila]
3PL.RLS-receive MUT.gold DAT MUT.advisor
'They received gold from the advisor' [B350]

c. ʔu-fa-cibo zexula [ba **z**umo]
1SG.RLS-DYN-throw MUT.coconut LOC MUT.well
'I threw the coconut into the well.' [B229]

Second, genitive nominals consistently surface with mutation. As shown in (20a,20b), the possessors *ohi* ('coconut tree') and *banua* ('village') surface as *nohi* and *ʙanua*, respectively.

(20) a. bulu **n**ohi
leaf MUT.coconut.tree
'Leaf of coconut tree' [B348]

b. ʔono **ʙ**anua
child MUT.village
'Child of the village' [B374]

Third, in experiencer constructions, as in (21), both experiencer and stimulus must be marked.

(21) ʔa-taʔu [**ʙ**aʔe]$_{EXP}$ [**n**ono matua]$_{STIM}$
ST-fear MUT.monkey MUT.child male
'The monkey is afraid of the boy' [B344]

On the preceding hypothesis, where argument marking is the realization of [ABS], the above instances of argument marking are unexpected. Assuming that in (19a,19b,19c) the prepositions *xö* and *ba* assign an oblique case to their complement, i.e. [DAT] and [LOC] respectively, then these arguments are expected to surface with a morphologically distinct case form than the one observed with prototypical absolutive arguments (which are presumably ABS-marked). Similarly, in the case of the possessors in (20a,20b), these nominals are expected to have a morphologically distinct form given that such arguments typically bear [GEN] and not [ABS]. Finally, in the case of the experiencer construction in (21), both subject and object share the same morphological form regarding argument marking, despite the subject not being a prototypical absolutive argument.

Given that the distribution of argument marking extends beyond prototypical absolutive arguments to obliques, genitives, and experiencer subjects, I argue that approaches that posit Nias exhibits an underlying absolutive case system must also posit that Nias exhibits an implausibly high degree of case syncretism. Namely, a syncretism between all cases except for ERG, i.e. ABS=OBL=GEN. Although case syncretism is well-attested in ergative-absolutive case systems (Dixon 1994; Zompì 2017; 2019; Smith et al. 2019, *i.a.*), the key issue is that for Nias this requires positing syncretism between all cases except for ERG.

The problem with positing this kind of syncretism is the following. Given that ABS-marking among core arguments has a wider distribution than ERG-marked core arguments in Nias (intransitive subjects and objects *vs.* transitive subjects), ABS is distributionally less marked than ERG (Dixon 1994). Moreover, it has been observed cross-linguistically that among syncretism patterns between core and peripheral cases, i.e. syncretism between ABS/ERG and cases that surface on obliques

(e.g. LOC), the peripheral case will be syncretic with the distributionally marked case. That is, in absolutive/ergative systems the peripheral case(s) will be syncretic with ERG (Blake 1994; Zompì 2017; 2019; Smith et al. 2019). However, in Nias this type of syncretism is not what is observed. Instead, on the case approach the key posit is that all peripheral cases are syncretic with the distributionally unmarked ABS case form. In this respect, Nias' argument marking pattern is rendered a typological outlier with respect to attested possible case syncretism patterns (in addition to displaying an otherwise unattested 'marked absolutive' case pattern).[4]

Given the above facts, I conclude that the distribution of mutated and unmutated arguments, quite generally, does not correspond to grammatical role. First, arguments which are presumably absolutive, i.e. (13-15b), (17a,18a), consistently fail to undergo mutation when dislocated to a clause-initial position. Second, in several contexts, non-absolutive arguments, i.e. oblique (19a,19b,19c), genitive (20a,20b), and subject experiencer (21) arguments, consistently undergo mutation. Tab. 2 summarizes these facts.

CONTEXT	MUTATION
Intransitive subjects	✓
Direct objects	✓
Oblique arguments	✓
Genitive arguments	✓
Experiencer subjects	✓
Transitive subjects	✗
Promoted subjects in passives	✗
Clause-initial nominals	✗

Table 2: Argument Marking Environments.

1.3 Challenges for mutation as dependent case

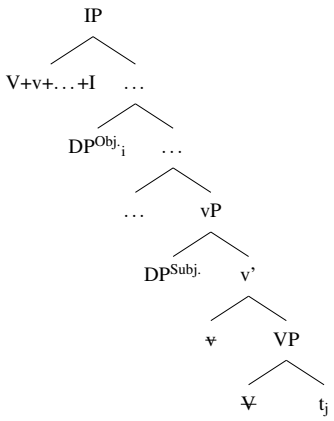
A core feature of the dependent case approach to argument marking is that the distribution of ABS-marking in Nias is regulated by whether a nominal c-commands another nominal in the syntax (Baker 2015). However, further examination of Nias' clausal syntax indicates that whether a nominal is marked (or not) is independent of whether that nominal c-commands another. In this section, I show that Nias' VOS order is derived through phrasal vP-fronting (rather than V-to-X movement, cf. Baker 2015). Consequently, I show that in transitive clauses, the subject and the object systematically do not stand in a c-command relation with each other and, thus, per the dependent case rule in (10) are wrongly predicted to both be realized as ABS-marked.

Previous research on V-initial languages propose that such orders are derived either by V-to-X movement (Koopman 1984; Travis 1984; Guilfoyle et al. 1992; Clemens & Polinsky 2017, *i.a.*) or by predicate vP/VP-fronting (Massam & Smallwood 1997; Massam 2001; Chung 2005; Coon 2010; Clemens & Polinsky 2017, *i.a.*). On the V-to-X analysis, Nias' VOS order would be derived as in (22), where V undergoes head-movement to a higher functional head in the inflectional domain, e.g. I^0, and the object DP undergoes movement to a position above the subject DP (or the subject DP projects as a right-side Spec of IP, as in Guilfoyle et al. 1992). On the vP-fronting approach, as in (23), the subject DP raises out of the predicate phrase and the remnant vP fronts

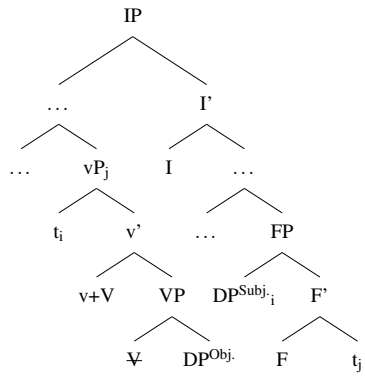
[4] Note, that on an approach to case that assumes a case containment hierarchy, where ABS⊂ERG⊂OBL, positing an ABS=OBL syncretism is an instance of an ABA syncretism pattern (in the sense of Bobaljik 2012), which is predicted to, quite generally, be an impossible pattern (Zompì 2019).

to a Spec, above the subject DP, i.e. Spec,IP, pied-piping the *in situ* object DP with it (see Cole & Hermon 2008 for an analysis of this type for Toba Batak).[5]

(22)



(23)



I argue that evidence from surface order of vP-internal elements indicates that Nias' VOS order is derived by phrasal movement, i.e. (23).

As shown in (24a,24b), frequency adverbs must precede the verb in the clause-initial position.

(24) a. **to?ölö** la-agö xö dʳa-Gusti
usually 3PL.RLS-stay DAT MUT.COL-Gusti
'Usually they stay with Gusti's family' [B484]

b. **ʔasese** la-fake gorokoro ʔira-ina meföna
often 3PL.RLS-use MUT.scoop ASS-mother in.early.days
'In the early days, women often used scoops' [B485]

Assuming that such adverbs are adjoined/base-generated low in the verbal field (Jackendoff 1972; Alexiadou 1997; Cinque 1999), the fact that they must surface clause-initially, preceding the main verb, indicates phrasal movement. In particular, on a phrasal movement approach, the above facts are easily accounted for in terms of extended verbal projection, i.e. vP, undergoing fronting where such elements, e.g. adjuncts, are pied-piped (Chung 2005; Clemens & Polinsky 2017). In contrast, on a V-to-X approach, such elements should be stranded lower in the structure, and, thus, should not surface clause-initially preceding the main verb.

Similarly, as (25a,25b) show, directional particles must surface adjacent to the main verb.

(25) a. ma=aekxu **tou** viga
PERF=fall down MUT.plate
'All the plates fell down' [B479]

b. möi **lawa** βöli z=o-guna
go up MUT.price REL=HAVE-need
'The prices of goods are going up.' [B150]

Following the assumption that directional particles are base-generated vP-internally (Emonds 1973; Neeleman & Weerman 1993; Ramchand & Svenonius 2002; Drummond 2023), the fact that such particles are adjacent to the clause-initial V, indicates that V-initial order is the result of phrasal

[5] As (23) illustrates, I assume that V quite generally undergoes head-movement to v in Nias. Additionally, for expository purposes, I have collapsed VoiceP and vP into a single projection (i.e. vP). While it may be the case that the Nias verbal field involves the projection of a VoiceP (cf. Toba Batak Cole & Hermon 2008), whether VoiceP projects does not make a material difference to the present discussion.

movement. If V-initial order were due to V-to-X movement, then the directional particles should be stranded clause-finally in their base-generation position inside the vP-remnant.

Given that Nias' VOS order is due to subject raising plus fronting of the vP-remnant, I argue that this poses a challenge for an approach that analyzes Nias' argument marking in terms of dependent case. As previously discussed, on this approach a DP is assigned ABS in contexts where that DP fails to c-command another DP within the case assignment domain (i.e. the IP domain). However, if underlying structure of Nias' transitive clause is derived as in (23), then neither argument stands in a c-command relation with the other. Thus, per the case assignment rule in (10), both arguments should be assigned ABS and surface marked (contrary to fact).[6]

A further issue with approaches that treat Nias' argument marking pattern in terms of dependent case comes from the argument marking observed with passive constructions (e.g. 18a) (and repeated below).

(18a) ba?a ni-fönu-i=nia ?idanö
tub PASS-fill-TR=3SG.POSS water
The tub which she filled with water. [B554]

As previously discussed, in the Nias passive the promoted subject surfaces clause-initially and, crucially, is unmarked. However, per the dependent case rule in (10), if an argument does not c-command another argument, then it is assigned ABS. Assuming that the passive in (18a) involves valency reduction, and that *ba?a* ('tub') is the sole argument, then per the dependent case analysis, the promoted subject should be ABS-marked. However, the promoted subject surfaces unmarked.[7]

Thus, the key issue for a dependent case approach to the distribution of argument marking in Nias is that such an approach predicts a tight correlation between the presence/absence of argument marking and (anti)-c-command. However, as shown this correlation fails to hold. Given that the distribution of argument marking does not depend on whether (or not) the argument stands in a c-command relation to another argument, this indicates that the distribution of argument marking is not a reflex of dependent case assignment.

1.4 Summary

To summarize, this section showed that the distribution of marked and unmarked arguments, quite generally, does not clearly correlate with the grammatical role of the argument. First, argument marking occurs on a number of arguments that are not traditionally marked with ABS, e.g. oblique arguments. Second, clause-initial arguments consistently fail to be marked, even when the argument is an absolutive. On an approach where argument marking is regulated by the presence of ABS (and the absence of marking with ERG), these facts are highly unexpected. That is, the distribution of marked arguments does track with the prototypical distribution of ABS-marking observed cross-linguistically. Further, on an approach where Nias argument marking is due to a dependent case rule (as in Baker 2015), these issues are further compounded due to such an account requiring all marked arguments in Nias to not only correlate with absolutive grammatical role but that such arguments must not c-command other arguments, which is not the case in Nias. Given that the

[6] One could assume (and departing from Baker 2015) that in Nias transitive constructions, the dependent case rule applies prior to the subject DP raising out of Spec,vP (and vP-fronting) such that the subject DP c-commands the object DP from Spec,vP, and, thus, bleeds ABS-assignment to the subject DP in transitives. However, this proposal would require an additional parameterization of the case algorithm, where vP is the dependent case assignment domain for ABS/ERG in Nias, but in all other languages, e.g. Warlpiri, IP is the dependent case assignment domain for ABS/ERG. Further, it has been proposed that vP is the dependent case assignment domain for DAT (Baker 2015; Yuan 2020). If this is the case, then Nias' case algorithm would require further parameterization, such that the ABS/ERG and DAT-assigning case domains do not coincide with both being the vP.

[7] Note that in (18a) the nominal *?idanö* is not an argument but rather is an instrumental adjunct, which quite generally are unmarked in Nias (Brown 2001).

distribution of argument marking does not correlate with grammatical role and, thus, fails to support the hypothesis that mutation is the realization 'marked absolutive' case system, the main analytical task then is two-fold: (i) define the domains that regulate where argument marking occurs; (ii) understand the grammatical processes that underlie mutation.

2 New analysis

The previous section showed that while in a restricted range of contexts morphological marking correlates with absolutivity, this correlation does not hold in the general case. In this section, I develop a new analysis of Nias' argument marking, which does not depend on grammatical role. The starting point for this analysis comes from the novel generalization regarding the distribution of marked arguments. Namely, that whether an argument surfaces as marked (or not) in Nias is conditioned by whether (or not) the DP is spelled-out in the same phase as a lexical head.

2.1 The syntax of argument marking

As was shown in the previous section, the correlation between the presence/absence of argument marking and grammatical role systematically breaks down in a well-defined range of cases. In this section, I will argue that the correct generalization that captures the distribution of marked arguments is the following:

(26) If a DP is in the same phase as a lexical X^0 at Spell-out, then the initial segment of the DP is marked with MUT.

Before showing that the generalization in (26) captures the contexts where mutation occurs, I will outline some basic architectural assumptions regarding the timing of transfer to the interfaces, the size of Spell-out domains, and which phrases constitute phases. Following previous works (Uriagereka 1999; Chomsky 2001, *i.a.*), I assume that syntactic structure is sent to the interfaces in 'chunks', which correspond to syntactic phases. Specifically, I adopt the assumption that a phase is complete upon the merger of a higher phase head (Chomsky 2001), and that upon the completion of the phase the full phase spelled-out to the interfaces (Fox & Pesetsky 2005; Bošković 2014; 2016; 2025). Following previous work in Distributed Morphology (Halle & Marantz 1993; Embick & Noyer 2007; Embick 2010, *i.a.*), I assume that at Spell-out syntactic terminals are associated with Vocabulary Items (i.e. the phonological realization of the terminal), which then feed into the module where phonological operations apply. The relevant processes, are illustrated in (27).

(27) $[_{XP}\ X_{\text{ph. head}} \dots [_{YP\text{ph./s.o. domain}}\ (ZP)\ [_{Y'}\ Y_{\text{ph. head}} \dots]]] \rightarrow [_{XP}\ X_{\text{ph. head}} \dots [_{YP}/\alpha \dots \beta/]]$

As for which phrases count as phases, I adopt the proposal developed in Bošković 2014 that the highest phrase in the extended domain of a lexical head delimits a phase domain. Concretely, on this approach (and what is relevant here) the highest phrases in the extended domain for every lexical head in Nias, i.e. V, P, N, delimits a phase domain.[8,9] Given the above assumptions, the key point regarding (26), is that it is a generalization that holds when the DP is spelled-out, i.e. upon the merger of the next higher phase head.

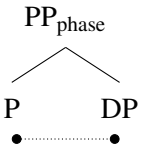
[8] While Bošković 2014 includes lexical adjective heads, i.e. A^0, as heads that have a phase domain and, thus, are phase delimiting, A^0 is omitted from the present discussion. As discussed in Brown 2001, it is not clear whether Nias instantiates A^0 as a lexical category (see also Baker 2003 and Chung 2012 for discussion on this issue regarding the status of lexical A^0 in Austronesian).

[9] It should be noted that there is a question regarding the phasal status of CP on this approach. Following a proposal in Bošković 2014, I assume that CP is a phase due to CP being the highest projection in general (in the clause) (although see Bošković 2014 for alternative proposals regarding how CP receives its phasal status).

Consider now the distribution of marked arguments in Nias. As was shown, oblique arguments, e.g. dative arguments (19a) are always marked. Assuming that the nominal is generated inside a PP as a complement to the preposition *xö*, as in (28), then argument marking correlates with the DP inside the PP phase when the structure is sent to PF where it is marked with MUT.

(19a) ʔi-beʔe gefe [$_{PP}$ xö **z**odʳöröu] ʔama Dali
3SG.RLS-give MUT.money DAT MUT.healer Ama Dali
'Ama Dali gave money to the village healer.' [B269]

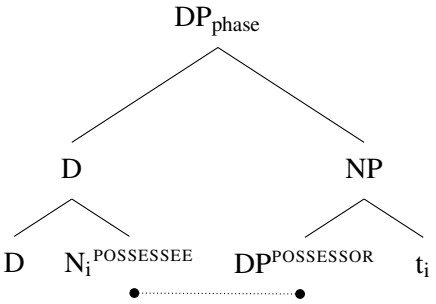
(28)



Similarly, in possessive constructions, the possessor is always marked, as in (20a). I adopt the assumption that Nias' possessive constructions involve the structure in (29) where POSSESSEE-POSSESSOR order is due to the POSSESSEE undergoing N-to-D movement above the POSSESSOR DP (cf. Ritter 1988; Duffield 1995; Borsley et al. 2007, *i.a.*). Subsequently, when the structure in (29) is spelled-out, the POSSESSOR will be inside the DP with the POSSESSEE, which contains a lexical N, thus, conforming to the generalization in (26).

(20a) bulu **n**ohi
leaf MUT.coconut.tree
'Leaf of coconut tree' [B348]

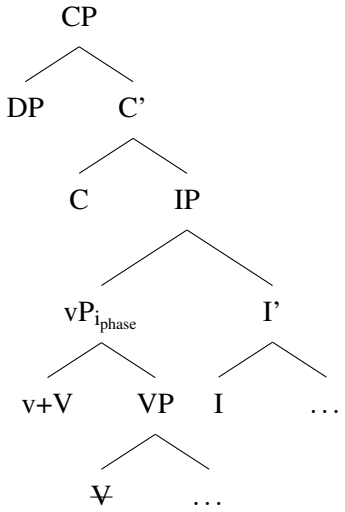
(29)



Consider now the cases where a nominal surfaces in a clause-initial position and is unmarked, e.g. topicalization (13). In such cases, the nominal is located in a clause-initial position, e.g. Spec,CP, as illustrated in (30). Importantly, in this position, the nominal is not in the same phase as the lexical V, which is inside the vP phase. Thus, when the clause-initial nominal is spelled-out in the CP phase, (26) will not hold, and the nominal will be unmarked.

(13) siʔo höʔö ma=i-taruʔ-ö ba danö
stick DIST.DEM PERF=3SG.RLS-plant-TR LOC MUT.ground
'That stick, he planted in the ground.' [B361]

(30)



A similar effect is observed in Nias passive constructions. As (17a) shows, the promoted subject surfaces in a clause-initial position and is unmarked.

(17a) ʔahe βatö ni-tefe=gu ʔidanö
floor PASS-sprinkle=1SG.POSS water
'The floor that I sprinkled with water.' [B553]

Further, in cases of object extraction, e.g. with *wh*-movement as in (31a,31b), the object is clause-initial and the verb must be marked with the passive morpheme.[10]

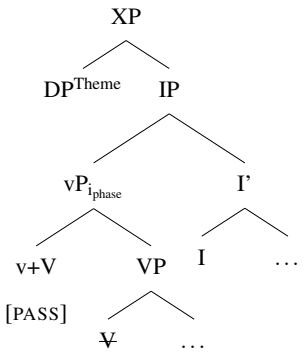
(31) a. hata ni-base-ʔö-u?
who PASS-wait-TR-2SG.POSS
'Who are you waiting for?' [B255]

b. haija ni-waö-nia xö-u?
what PASS-say-3SG.POSS DAT-2SG.POSS
'What did he say to you?' [B350]

That passivization is required for object extraction and that promoted subjects quite generally surface clause-initially, suggests that promoted subjects are external to the vP-phase. Given that promoted subjects are vP-external, they are spelled-out in a domain without a lexical head and, thus, surface unmarked (32).

[10] In this respect, Nias displays an Ā-extraction asymmetry that is typologically common in Austronesian. Namely, that only logical subjects can undergo Ā-movement from their base position, while logical objects must first undergo an additional operation, e.g. passivization, that feeds Ā-movement (Keenan & Comrie 1977, *i.a.*). Further, this type of extraction asymmetry can also be taken to be additional evidence that Nias' VOS order is due to vP-fronting (Chung 2005; Cole & Hermon 2008), Assuming that a moved vP is an island for extraction (Wexler & Culicover 1980), then the inability of Nias' objects to be extracted without an additional operation, e.g. passivization, straightforwardly follows given that the object is located inside the vP island at the point in which Ā-extraction would occur. Conversely, if Nias' VOS order were due to V-to-X movement, then Ā-extraction of both subject and object should be possible without additional operations (due to the absence of vP-fronting).

(32)



To summarize, in the case of obliques, genitives, clause-initial dislocated arguments, and promoted subjects, the generalization in (26) clearly tracks when an argument will be marked with mutation and when it will surface unmarked. In cases where the DP is spelled-out in the same domain as a lexical head, the nominal is marked. But when the DP is spelled-out in a domain without a lexical head, it is unmarked.

2.1.1 Deriving the core argument pattern

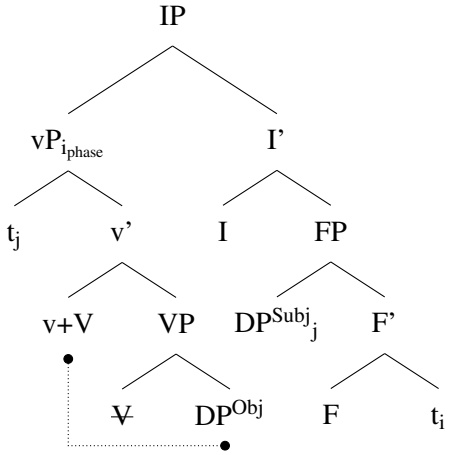
As was shown, intransitive subjects (5b) and objects (6a) are consistently marked to the exclusion of transitive subjects (6a), which surface unmarked.

(5b) ma=aso?a **d**uhituhi
PERF=fall MUT.trivet
'The trivet stone fell.' [B559]

(6a) ma=i-uri **z**awi Fasui.
PERF=3SG.RLS-keep MUT.cattle Fasui
'Fasui kept cattle.' [B366]

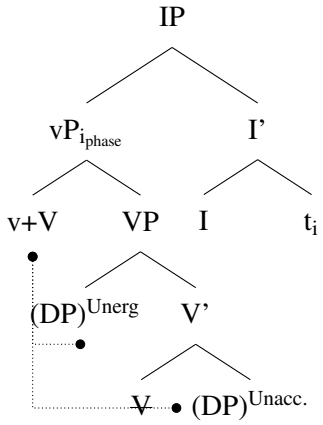
Consider first transitive constructions, and how the structure of such clauses bears on the generalization developed in (26). As argued for in §1.3, the Nias transitive clause involves the subject DP raising to a clause-medial position, i.e. Spec,FP, out of the vP and the vP-remnant fronting with the object remaining vP-internal, as in (33). Importantly, this means that while the object remains within the vP phase, the subject is located outside of the vP phase in Spec,FP. Consequently, the subject is not located within the same phase as the lexical V. Thus, given the structure of transitive clauses the arguments adhere to the generalization in (26) where the object surfaces marked but the subject is unmarked.

(33)



Consider now intransitive constructions, as in (5b), where the subject always surfaces marked. I argue that intransitive subjects uniformly remain within the vP phase, which I provide independent evidence for below. Because of this, they are always marked, as per the generalization in (26). Thus, I propose that the intransitive clause has the structure in (34):

(34)



Evidence that intransitive subjects are uniformly structurally lower than transitive subjects comes from causativization patterns. As shown, when an unaccusative predicate, e.g. a stative as in (35b), is marked with the morphological causative *f(a/e)-*, the theme argument is retained and an agent/causer argument is introduced.

(35) a. ʔa-kao ʔita
ST-have.difficulty MUT.1PL.INCL
'We are having difficulties' [B236]

b. ʔi-**f**-a-kao ʔita
3SG-CAUS-ST-have.difficulty MUT.1PL.INCL
'He is causing us difficulty' [B236]

In contrast, when a transitive predicate is causativized, as in (36b), a causer subject is introduced and the causee (i.e. demoted subject) must surface as an oblique argument.

(36) a. ʔu-sura zura *pro*
1SG-write MUT.letter
'I wrote a letter' [B259]

b. ʔi-**fa**-sura-ʔö zura xö-gu fandrita
3SG-CAUS-write-APP MUT.letter DAT-1SG.POSS priest
'The priest got me to write a letter' [B259]

In the case of causatives of unergatives, the realization of the intransitive subject patterns with the unaccusative in (35b). As shown in (37b), the causee (i.e. the unergative subject) is not realized as an oblique argument (as with transitive subjects) and no argument demotion is visible.

(37) a. ʔaʔege dʳaga
laugh MUT.1PL.EXCL
'We laughed' [B236]

b. ʔi-**f**-aʔege dʳaga baʔe
3SG-CAUS-laugh MUT.1PL.EXCL monkey
'The monkey made us laugh' [B236]

I interpret the above pattern as indicating that subjects of unergatives are base-generated lower than transitive subjects, which are base-generated in Spec,vP (Massam 2009; Tollan & Oxford 2018; Kouneli 2021). Thus, since vP uniformly lacks an argument merged into its Spec in intransitives, when such predicates are causativized the causer can merge to Spec,vP without requiring subject demotion— i.e. the causee being realized as an oblique.

Further data that indicates this structural difference between intransitive and transitive subjects concerns differences regarding these arguments associating with quantificational elements. In Nias, the quantificational adverb *ʔoi* can induce a quantificational interpretation of an argument that is within its scope, as in (38) (Brown 2001). As (38a,38b) show, *ʔoi* can associate with both unergative and unaccusative subjects, inducing a quantificational interpretation.[11] But in transitive constructions, as in (38c), while *ʔoi* can associate with an object, it cannot with the subject (Brown 2005).

(38) a. M=oi mo-gamagama niha
PERF=all DYN-armament MUT.person
'Every man wore weapons/was armed' [B202]

b. ʔoi to-kia dʳaga
all RES-shock MUT.1P.EXCL
'We all were shocked.' [B493]

c. ʔoi dʳa-mbe v-a-nolo-ra xö dʳa-ono
all 3PL.IRR-give MUT.NMLZ-IMPV-help DAT MUT.COL-child
'They will give all of their help to the children.' [B492]

I interpret this split between transitive and intransitive subjects as indicating that transitive subjects are located in a position that is outside *ʔoi*'s domain, and, thus, cannot associate. Conversely, given that intransitive subjects can associate with *ʔoi*, this indicates that intransitive subjects are in a position below *ʔoi*. Concretely, following the analysis of quantificational particles in Clemens 2019 and Tollan & Massam 2022, I propose that *ʔoi* heads a phrase, which can associate with an argument in its c-command domain. In Nias, I suggest that this phrase merges low to the vP. Evidence that *ʔoi* merges low comes from scope data. As (39) shows, when negation and *ʔoi* co-occur, *ʔoi* scopes below negation. Assuming that NegP projects low in the structure, within the extended verbal projection, this indicates that *ʔoi* must merge to a phrase below NegP—i.e. vP.

(39) löna ʔoi man-a ira
NEG all IMPV-eat MUT.3PL
'Not all of them ate.' [B493]

Thus, given the causativization and interpretative facts regarding *ʔoi*, I conclude that while transitive subjects are vP-external, intransitive subjects are uniformly vP-internal in Nias.[12]

[11] Note that while Nias does not clearly distinguish between unaccusative and unergatives predicates syntactically, it does distinguish between predicates with an agentive vs. non-agentive subject morphologically. In (38a), the stem is marked with the 'dynamic' prefix *mo-*, which derives a intransitive agentive verb from the stem—i.e. a prototypical unergative verb. In (38b) the stem is marked with the 'resultative' prefix *to-*, which derives a 'resultant state' verb from the stem—i.e. a prototypical unaccusative verb (see Brown 2001: 196-207, 210-216, for discussion).

[12] As for why intransitive subjects remain vP-internal (unlike transitive subjects), this is an open question. One reason may be due to an anti-locality effect, where a phrase moving from Spec,XP must cross a maximal projection distinct from XP (Deal 2019; Erlewine 2020). Assuming that vP delimits a phase and that movement through a phase boundary requires an intermediate movement step at the phase edge (Chomsky 2001), then all movement out of vP requires Spec,vP as an intermediate landing site. However, if unergative subjects are base-generated in Spec,VP, then movement outside of vP would require Spec,VP-to-Spec,vP movement, which would be blocked due to anti-locality— trapping unergative subjects inside vP. Crucially, though this explanation would only apply to unergative subjects since for unaccusative

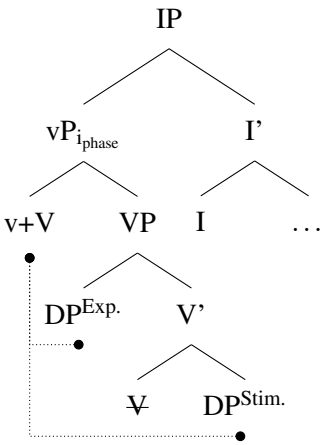
As was shown in §1.2, there is an exceptional case regarding the absence of marking in transitive subjects, which concerned experiencers. Unlike agentive transitive subjects, experiencer subjects were shown to consistently surface marked, as in (21) (repeated below). Interestingly, unlike agentive transitive subjects, experiencers can also associate with *oi*, as shown in (40).

(21) ʔa-taʔu **β**aʔe **n**ono matua
ST-fear MUT.monkey MUT.child male
'The monkey is afraid of the boy' [B344]

(40) ʔoi ʔomasi **d^{r}**aga **n**asu
all like MUT.1P.EXCL MUT.dog
'We all like dogs.' [B491]

Given the above facts in (21) and (40), following Belletti & Rizzi (1988) among others, I argue that experiencer subjects are base-generated lower in the structure than agentive transitive subjects. Moreover, given that experiencer subjects consistently surface marked and can associate with *ʔoi*, these subjects remain vP-internal and are spelled out in the same phase domain as V (as with intransitive subjects), as sketched in (41). Thus, given the structure in (41), experiencer constructions adhere to the generalization in (26).[13]

(41)



Zooming out, this section demonstrated that the argument marking pattern in Nias is captured by the generalization in (26). Crucially, this generalization does not make reference to the grammatical role of the argument. Rather, I argued that the relevant factor regulating argument marking is the structural position of the argument at PF. When the DP is spelled-out in the same phase domain as a lexical head, it surfaces marked; when it is spelled-out in phase distinct from a lexical head it is unmarked. Moreover, I showed that this generalization can capture the argument marking pattern observed with core arguments. Specifically, I showed that the apparent 'marked absolutive' surface pattern is due to a systematic difference in structural position between intransitive subjects and objects, and agentive transitive subjects.

subjects, which are base-generated in CompVP, anti-locality wouldn't apply. Interestingly, if all intransitive subjects are base-generated in the same Spec (as proposed for Kipsigis in Kouneli 2021), e.g. Spec,VP, then movement outside of vP would be uniformly blocked for intransitive subjects.

[13] Note that on the dependent case approach in Baker 2015, Nias' experiencer subjects are posited to be merged inside a covert PP-shell that assigns [OBL] to the experiencer and blocks it from c-commanding the stimuli. The present approach, however, does not require stipulating additional silent structure in this way, and experiencers simply merge as DPs, as in (41).

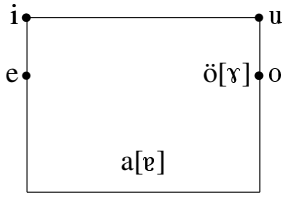
2.2 Argument marking as a morpho-phonological process

As argued in §2.1 the distribution of marked arguments is regulated by the structural position of the DP at Spell-out to PF. Namely, whether the DP is spelled out in the same phase as a lexical head. In this section, I argue that the surface form of argument marking the result of a morpho-phonological process. After outlining basic elements of Nias phonology in §2.2.1, I show that the surface form of marked or 'mutated' arguments is Nias is phonologically predictable. Further, a phonological analysis of Nias' mutation pattern is developed, illustrating that mutation can be analyzed in such terms. Additionally, in §2.2.3 I discuss Nias' 'mutated pronouns', where I argue that this alternation is a distinct type of alternation from argument marking on lexical nouns but nonetheless constrained by phase-based locality. Moreover, I show that that the alternation that Nias' pronouns participate in is not regulated by grammatical role, and, thus, further evidence against analyzing Nias' mutation patterns as the realization of case.

2.2.1 Phonology background

As shown are Nias' vowel (42) and consonant (43) inventories. As discussed in Brown (2001; 2005) the Nias syllable adheres to a (C)V template (where V may be bimoraic).

(42) Vowel Inventory



(43) Consonant Inventory

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Platal	Velar	Glottal
Plosive	b		t d				k g		'[ʔ]
Plosive, Trilled Release			ndr[d^r]						
Nasal	m		n						
Trill	mb[β]		r						
Fricative		f v	s					kh[x]	h
Affricates				c[ʧ] z[ʤ]					
Approx.		β[ʋ]					y[j]	w	
Lat. Approx.			l						

While medial syllables may lack an onset, Nias has a strict ban on initial syllables being onsetless, i.e. no V-initial words (*[$_\omega$V). In such cases, where the underlying initial syllable lacks an onset, a glottal stop [ʔ] is inserted as a default onset (44) (Brown 2001). This process is illustrated in Tab. 3. As shown in Tab. 3, when the V-initial stem is prefixed with the intransitive *m-*, no glottal stop appears on the initial segment of the stem, e.g. [m-uta] ('to vomit'), cf. *[m-ʔuta]. But in contexts where the stem is the initial element, i.e. with the transitivizing suffix *-iö*, a glottal stop surfaces as the initial element [ʔuta-iö] ('to throw up *x*') (cf. *[uta-iö]).[14]

(44) **ʔ-Insertion:** $\emptyset \rightarrow$ ʔ / _[$_\omega$V (Brown 2005: 41)

[14] Note that while the Nias orthography in Donohue & Brown 1999; Brown 2001; 2005 omits word-initial glottal stops, I include them in the examples in this paper to more transparently reflect phonological form.

Stem	Intransitive	Transitive
-uta ‘to vomit’	m-uta	ʔuta-iö
-eʔe ‘to cry’	m-eʔe	ʔeʔe-si

Table 3: Glottal Insertion.

2.2.2 Mutation as nasal insertion

As shown in Tab. 4, in the case of consonant-initial nominals, argument marking is realized in two grades. In cases where the initial consonant is voiceless, e.g. *faxe*, argument marking is realized as voicing of the initial segment, e.g. *vaxe*. In cases where the initial consonant is a voiced obstruent, e.g. *doi*, argument marking is realized as trilling *dʳoi*. In cases where the initial segment is a sonorant, e.g. the nasal initial *nagole*, argument marking does not occur even in contexts where arguments typically surface marked. As (45) illustrates, the object is nasal-initial and surfaces unmarked.

Class	Unmutated	Mutated	Alternation
-VOICE		**Voicing**	
	faxe ’rice’	vaxe	f ∼ v
	tanö ‘land’	danö	t ∼ d
	siʔo ‘stick’	ziʔo	s ∼ z[ʤ]
	ciʔaciʔa ‘gecko’	ziʔaciʔa	c[ʧ] ∼ z[ʤ]
	kefe ‘money’	gefe	k ∼ g
+VOICE,-SONORANT		**Trilling**	
	baβi ’pig’	βaβi	b ∼ β
	doi ‘thorn’	dʳoi	d ∼ dʳ
+SONORANT		**No Mutation**	
	Mili ‘Mili (name)’	-	-
	nagole ‘meat’	-	-
	labalaba ‘large spider’	-	-
	rake ‘coral’	-	-
	dʳi ‘mosquito’	-	-
Misc.		**No Mutation**	
	Gomo ‘Gomo (name)’	-	-
	xö ‘possessions’	-	-

Table 4: Consonant Mutation.

(45) ʔu-taba nagole faoma balatu
1SG.RLS-cut.up meat with knife
‘I cut the meat with a knife’ [B361]

In the case of vowel-initial nominals, the pattern is somewhat different. As shown in Tab. 5, the surface form of marked vowel-initial nominals appears to be divided into two classes. Nominals where argument marking is realized as a prothetic [n] and nominals where it is realized as a prothetic [g]. In cases where the nominal is unmarked, it surfaces with a prothetic [ʔ], per the glottal-insertion rule in (44).

Further, as observed in Brown 2001, there are several cases where the marked forms of vowel-initial nominals are seemingly not predictable on the basis of their unmarked surface forms. For example, as Tab. 5 shows, [ʔöri] ‘federation’ and [ʔöri] ‘amulet’ are only phonologically

Unmutated	Mutated
***n*-mutation**	
ʔöri 'federation'	nöri
ʔuβu 'plank'	nuβu
ʔiβa 'sibling'	niβa
ʔete 'bridge'	nete
ʔadu 'statue of ancestor'	nadu
***g*-mutation**	
ʔöri 'amulet'	göri
ʔuβu 'part of coconut'	guβu
ʔiβö 'movement'	giβö
ʔeteʔete 'long wave'	geteʔete
ʔadulo 'egg'	gadulo

Table 5: Vowel Mutation.

distinguished in their marked forms: [nöri] and [göri], respectively. Subsequently, Brown (2001) interprets this observation as indicating that argument marking for vowel-initial nominals is not predictable and that the marked form of such nominals are lexically specified. Although Brown's (2001) observation on this point may suggest that the marked forms of vowel-initial nominals are not phonologically predictable, I argue that this division regarding the surface form of marked vowel-initial nominals is, in fact, predictable on the basis of the underlying form of the nominal stem, which is obscured by the surface forms due to glottal-insertion. I propose that for vowel-initial stems, the surface form of marked nominals uniformly involves a prothetic [n]. In cases, where the vowel-initial stem is unmarked, glottal-insertion (i.e. (44)) applies and the nominal surfaces with a prothetic [ʔ]. In cases where the surface form of the marked nominal involves an initial [g], the underlying form of the nominal stem involves an initial [ʔ], which is visible in contexts where the nominal surfaces unmarked. In contexts where [ʔ]-initial nominals are marked, the general pattern for voiceless consonant-initial nominals applies and the initial [ʔ] undergoes voicing surfacing as [g]. Thus, Nias' seeming 'unpredictable' vowel-initial pattern is an artifact of the glottal-insertion rule (with applies to vowel-initial forms) obscuring the difference between the underlying forms between the two classes of nominals.[15]

To summarize, I conclude that the surface forms of marked arguments are phonologically predictable on the basis of the underlying form of the nominal stem. In cases where the nominal is consonant-initial the nominal undergoes one of two grades of lenition and is either voiced or trilled. In the case of vowel-initial nominals, a prothetic [n] surfaces on the stem. Given this empirical situation, I propose that Nias' argument marking pattern is result of a phonological operation, which occurs in the argument marking contexts previously discussed. Specifically, that in such

[15] Interestingly, Brown (2001) gives diachronic evidence that supports the present hypothesis. As shown in (i.), stems that are marked with [g] are stems that had uvular stops (i.e. [q]) as the initial segment in Proto-Austronesian, while stems that are marked with [n] were vowel-initial.

(1) a. PA: *qateluR → Nias: gadulo 'egg'
b. PA *qalipan → Nias: galifa 'centipede'
c. PA: *aNak → Nias: nono 'child'
d. PA: *uRát → Nias: nuo 'vein/sinew' (Brown 2001: 71)

Given that Nias lacks uvular stops in its phonological inventory, a reasonable hypothesis is that a change has occurred where uvular stops where replaced with glottal stops for these forms.

contexts, a phonological operation is triggered where a prothetic nasal is inserted to the nominal, as in (46) (where *M* stands in for the marking context).

(46) $\emptyset \rightarrow \text{n} \; / \; M_[_{\omega}$

Given (46), the vowel-initial argument marking pattern (as in Tab. 5) follows straightforwardly from the application of (46), as shown in (47).

(47) **Vowel mutation:** *ohi→nohi* 'coconut tree'
  a. /ohi/
  b. **n**ohi — Output after *n*-insertion (46)

In the case of the argument marking pattern observed with consonant-initial segments, I argue that (46) applies in these contexts as well and a prothetic [n] is inserted to the stem. However, in such contexts, I argue that the insertion of [n] triggers lenition on the stem. Specifically, when the initial segment is a voiced obstruent, the segment becomes sonorant, which is realized as trilling (48a) and in contexts where the initial segment is voiceless, i.e. lacks a voice feature, the segment undergoes voicing (48b). For stems where the initial segment is a sonorant, neither (48a) or (48b) occurs.[16]

(48) a. $\begin{bmatrix} \text{+CONS} \\ \text{-SONORANT} \\ \text{+VOICE} \end{bmatrix} \rightarrow [\text{+SONORANT}] \; / \; [\text{+NASAL}]_$

b. $\begin{bmatrix} \text{+CONS} \\ \text{-SONORANT} \end{bmatrix} \rightarrow [\text{+VOICE}] \; / \; [\text{+NASAL}]_$

Finally, given that Nias quite generally disallows complex onsets, i.e. CC clusters, I argue that the repair rule in (49) is operative and the nasal deletes.

(49) ***CC resolution:** $\text{C} \rightarrow \emptyset \; / \; _\text{C}$

To illustrate how the above processes conspire to output Nias' argument marking pattern, consider two stems *faxe* and *bavi*. In the first case, as sketched in (50), (46) occurs and creates a context for the voicing process in (48b) to occur. Finally, given the prohibition on CC sequences, the nasal deletes, i.e. (49), generating the surface form *vaxe*. In the second case, as in (51), (46) occurs and creates a context for trilling (48a). Here as well, (49) then occurs and [n] deletes, thus, generating the surface form *ʙavi*.

(50) **Voicing Mutation:** *faxe→vaxe* 'rice'
  a. /faxe/
  b. nfaxe — *n*-insertion (46)
  c. nvaxe — Voicing (48b)
  d. **v**axe — CC resolution and output (49)

(51) **Trilling Mutation:** *bavi→ʙavi* 'pig'
  a. /bavi/
  b. nbavi — *n*-insertion (46)
  c. nʙavi — Trilling (48a)
  d. **ʙ**avi — CC resolution and output (49)

[16] Note that the for the segments [s] and [ʧ] voicing is realized as affrication, where the initial segment surfaces as [ʤ]. This may provide further evidence for the presence of an abstract nasal segment, which cross-linguistically has been observed to trigger affrication in such contexts (Steriade 1993).

Note as well, that that 'g-mutation' pattern in Tab. 5 is straightforwardly derived as an instance of voicing mutation. As illustrated in (52), with the stem *ʔöri*, after nasal insertion, the initial [ʔ] undergoes voicing to [g]. Then the [n] deletes, outputting the surface form: [göri].

(52) **'g-mutation':** *ʔöri→göri*

a. /ʔöri/

b. nʔöri — *n*-insertion (46)

c. ngöri — Voicing (48b)

d. **g**öri — CC resolution and output (49)

To summarize, I showed that Nias' mutation pattern is a phonologically predictable process. Specifically, in mutation contexts a prothetic nasal is inserted. In cases of a vowel-initial stem, the nasal surfaces. But in contexts where the stem is consonant-initial, the insertion of the nasal triggers lenition of the adjacent segment plus the deletion of the nasal.

Given the present analysis, Nias' mutation system displays a striking parallel to mutation systems observed cross-linguistically. For example, mutation in Welsh. Welsh exhibits a phonologically predictable mutation pattern, where voiceless segments undergo voicing and voiced stops undergo spirantization (Pyatt 1997; Roberts 1997; Tallerman 2006; Borsley et al. 2007, *i.a*). This phenomenon is illustrated in Tab. 6, where the 3SG.M possessive pronoun *ei* triggers mutation on the nominal complement (examples from Pyatt 1997).

Unmutated	Mutated	
pen	ei **b**en	'(his) head'
tad	ei **d**ad	'(his) father'
bara	ei fara [**v**ara]	'(his) bread'
mab	ei fab [**v**ab]	'(his) son'

Table 6: Welsh Soft Mutation.

Importantly, as with Nias' mutation system, Welsh mutation contexts do not strictly correlate with grammatical role(s). Rather, mutation occurs when the target is realized in a particular syntactic configuration with the mutation trigger. As illustrated in (53), in transitive VSO clauses, the direct object is c-commanded by the triggering element (i.e. the subject here) and the object undergoes soft mutation (i.e. *beic* ≈ *feic*). However, when the direct object is fronted and is not dominated by a trigger, as in (54), it surfaces unmutated.

(53) Prynodd y ddynes **f**eic.
buy.PST.3SG the woman bike
'The woman bought a bike' (Borsley et al. 2007: 224)

(54) Beic/*feic brynodd y ddynes.
bike buy.PST.3SG the woman
'It was a bike that the woman bought.' (Borsley et al. 2007: 233)

Additionally, elements that are non-arguments also may undergo mutation. As shown below, predicates of copular clauses (55) and non-finite verbs (56) may also be mutation targets when they are realized in mutation triggering syntactic contexts.

(55) Mae'r golchdy agosaf [PP **d**air milltir i ffwrdd]. (tair)
is-the launderette nearest three miles away
'The nearest launderette is three miles away.' (Tallerman 2006: 1769)

(56) Gall y dyn [$_{XP}$ **d**dreifio car]. (dreifio)
can the man drive car
'The man can drive a car.' (Roberts 1997: 159)

Thus, the key parallel between the distribution of mutation in Welsh and Nias, is that the relevant factor regulating mutation is not grammatical role, but whether or not a potential mutation target is realized in a specific triggering syntactic configuration. As was shown in §1, the environments where an argument mutates in Nias does not correlate with the grammatical role of the argument but rather the position that the nominal surfaces in. As with Nias, in Welsh when an element is realized in a triggering syntactic configuration it undergoes mutation; when it is realized in a non-triggering configuration it does not.

2.2.3 Pronoun Mutation

While argument marking on lexical nouns was shown to be predictable, Brown (2001; 2005) observes that pronouns participate in a distinct, non-phonologically predictable alternation—termed 'pronoun mutation'. While Brown (2001) identifies this alternation as tracking grammatical role—i.e. case, I argue that this alternation is best explained without reference to the grammatical role of the pronoun.

As shown in Tab. 7, Nias has two pronouns series which are described in Brown 2001; 2005 as 'unmutated' and 'mutated'.

	Unmutated	Mutated
1SG	ja?o	d^{r}ao
1SG.EMPH	ja ?oto	d^{r}aoto
2SG	ja?ugö	d^{r}augö
3SG	ja?ia	ja
1PL.INCL	ja?ita	?ita
1PL.EXCL	ja?aga	d^{r}aga
2PL	ja?ami	mi
3PL	ja?ira	?ira

Table 7: Pronoun Mutation.

The motivation for assimilating the two pronoun patterns with the mutation pattern observed with lexical nouns is due to the distribution of pronouns in the two series. Specifically, 'mutated' pronouns occur in the same contexts as marked lexical nouns, i.e. intransitive subjects (57a) and objects (57b) but do not surface as transitive subjects. Conversely, 'unmutated pronouns' consistently appear as transitive subjects (57b), as with unmarked lexical nouns.

(57) a. a-rörö **d^{r}ao**
ST-distract MUT.1SG
'I'm distracted' [B192]

b. ?i-bunu **ja** **ja?ia**.
3SG.RLS-kill MUT.3SG 3SG
'He killed him/himself.' [B542]

In this respect, the distribution of the two pronoun series appears at first glance to track with the distribution of marked and unmarked lexical nouns that are core arguments. Namely, that mutated pronouns pattern with marked core arguments in being absolutive arguments. While unmutated pronouns pattern with unmarked core arguments in being ergative arguments.

Despite the distributional similarity between (un)mutated pronouns and (un)marked core arguments, there are several important differences that indicate that the alternation observed with pronouns is not due to an underlying phonological process of the type observed with lexical nouns. First, as shown above in Tab. 7, while the initial segment of some pronouns appear to undergo the same lenition process characteristic of marking on lexical nouns, e.g. the approximate [j] undergoing trilling [dʳ] with the 1SG, in other contexts no lenition process occurs. For example, in the 3SG the initial [j] does not undergo lenition in the mutated case. Second, the surface forms of mutated pronouns suggest that pronoun mutation is a subtractive process. While the surface form of marked lexical nouns never involve phonological reduction of the stem, e.g. syllable deletion, pronoun mutation consistently involves this type of reduction. Throughout the entire mutated pronoun series the stems are consistently phonologically reduced compared to their unmutated counterparts, as in Tab. 7. Given the above differences between 'pronoun mutation' and argument marking on lexical nouns, I conclude that the alternation observed with pronouns in Nias cannot be the result of the same phonological process that is operative with argument marking on lexical nouns.

Note as well, that explaining the observation that 'mutation' on pronouns involves (morpho)-phonological reduction is especially challenging for an account that uniformly identifies mutated forms with ABS-case. If mutated forms are the realization of ABS—i.e. the morphologically marked case, then ABS-marked pronouns are expected to be more (morpho)-phonologically marked than their ERG, i.e. morphologically unmarked, counterparts. While this is the case for lexical nouns, in the case of pronouns the reverse is observed. Namely, that pronouns that are absolutives are (morpho)-phonologically less marked than pronouns that are ergative.

While the alternation observed with Nias' pronouns does not correspond to the alternation observed with (un)marked lexical nouns, Nias' pronoun alternation does appear to show behavior of characteristic of clitic and prosodically independent pronouns. As shown below, Nias' 'mutated' pronouns display properties that are characteristic of clitics (Cardinaletti & Starke 1999; van Riemsdijk 1999). As illustrated in (58a,58b), when pronouns occur in cleft constructions or marked with the focus particle *ha*, they surface in their unmutated forms. Conversely, cases of mutated pronouns occurring in contexts where they are foci are unattested in Brown 2001; 2005.

(58) a. ya?ia z=o=lau faya
3SG REL=IMPV=do lie
'It is HE who is the liar' [B444]

b. ha ja?ugö/ ?ö-fa-manömanö-si
only 2SG 2SG.RLS-DYN-talk-TR
'You're just talking about yourself' [B175]

That 'unmutated' pronouns may be focused, while 'mutated' pronouns occurring in focus contexts are unattested in Brown 2001; 2005 suggests that the difference between the two pronoun classes is one of prosodic strength. Namely, that while 'mutated' pronouns are prosodically deficient clitics, unmutated pronouns are prosodically independent (Cardinaletti & Starke 1999). Further, this difference in strength accounts for the distribution of 'mutated' pronouns being restricted to objects (57b) and intransitive subjects (57a). Concretely, I propose that at PF, a clitic (Cl^0), owing to its prosodic deficient status, attaches to a host within the spell-out domain. Given the syntactic structure of transitive (33) and intransitive (34) clauses, both the object and intransitive subject clitic are within vP domain, and, thus, can cliticize to V (as in (59)).

(59) a. $[_{vP}\ v^0{+}V^0\ Cl^0]$ Syntax
b. $(_{\text{S.O. Domain}}\ (_{\omega}\ \text{v+Verb}\ )\ (_{\sigma}\ \text{Cl}))$ Spell-out
c. $(_{\text{S.O. Domain}}\ (_{\omega}\ (_{\omega}\ \text{Verb}\ )\ (_{\sigma}\ \text{Cl})))$ Cliticization

Consider now transitive subject pronouns, which never surface as reduced 'mutated' pronouns, i.e. (57b). Given that transitive subjects, unlike objects and intransitive subjects, evacuate vP, they are no longer within the vP spell-out domain. Thus, cliticization on to V is impossible (as in (60)) and only a full, i.e. prosodically independent, pronoun may surface here.[17]

(60) a. $[_{IP}\ [_{vP}\ v^0{+}V^0\ DP]\ldots\ Cl^0]$ Syntax

b. $*(_{\text{S.O. Domain}}\ (_{\omega}\ \text{v+Verb})\ (_{\omega}\ DP\ ))\ (_{\text{S.O. Domain}}(_{\sigma}\ Cl\ ))$ Spell-out & No Cliticization

Interestingly, this account can explain the exceptional behavior of subject pronouns in experiencer constructions, which surface 'mutated', as in (61a,61b).

(61) a. ʔa-taʔu dʳao nasu
ST-fear MUT.1SG MUT.dog
'I'm afraid of the dog/dogs.' [B578]

b. ʔogoro dʳao ga-mua-ta-nia
disgust MUT.1SG MUT.IMPV-do-NMLZ-3SG.POSS
'I am disgusted by her behavior.' [B578]

As was shown in §2.1.1, experiencer subjects are base-generated below the vP and do not raise out. Due to experiencers remaining vP-internal, such arguments will be spelled out in the same domain as the verbal host. Thus, experiencers can be realized as clitic pronouns, which can attach to the verb at PF, as in (62).

(62) a. $[_{vP}\ v^0{+}V^0\ Cl^0{}_{EXP}\ DP_{STIM}\ ]$ Syntax

b. $(_{\text{S.O. Domain}}\ (_{\omega}\ \text{v+Verb}\ )\ (_{\sigma}\ Cl)\ (_{\sigma}\ DP))$ Spell-out

c. $(_{\text{S.O. Domain}}\ (_{\omega}\ (_{\omega}\ \text{v+Verb})\ (_{\sigma}\ Cl\ ))\ (_{\omega}\ DP\ ))$ Cliticization

To summarize, I conclude that the alternation observed regarding Nias' pronoun system is not the same as the mutation system observed with lexical nouns. Rather, in the case of pronouns the relevant difference is a difference in pronoun type—i.e. clitic vs. independent pronoun. Moreover, this alternation between clitic and independent pronoun is regulated by the position that the nominal is spelled-out in— i.e. whether it is spelled-out vP-internally or externally. In this way, Nias' pronoun alternations parallels the mutation on lexical nouns, where the underlying factor that regulates these alternations is not the grammatical role of the argument, but the structural position it is spelled-out in.

2.2.4 Syntactic and mutation domains

On the present analysis, the surface form of marked or 'mutated' arguments is due to the insertion of a nasal segment to a nominal stem. Moreover, it was shown that the relevant contexts where this process occurs is when a DP is in the same phase as a lexical head and is spelled-out, as stated in (26). Thus, in light of the present analysis of argument marking as nasal insertion, the generalization can be restated as follows:

(63) If a DP is in the same phase as a lexical X^0 at Spell-out, then the initial segment of the DP is marked with [n] at PF.

[17] As shown in Tab. 7 all unmutated pronouns have *ja* as the initial syllable of the stems. If it is the case that full pronouns are structurally more complex than clitics, and this additional structure results in prosodic independence (as in Cardinaletti & Starke 1999), then it may be the case that *ja* is the morphological realization of this additional structure.

Given the generalization in (63), I propose that the aforementioned phonological process that underlies argument marking is due to the application of (64) at PF. As shown, (64) states that nasal insertion occurs on a nominal in contexts where the nominal phrase is within the same phase (i.e. YP), as a lexical head (where X^0= V, P, N).

(64) $\emptyset \rightarrow$ n/ [$_{YP_{phase}}$ X^0 … _[$_{DP}$ …]]

Given the rule in (64), nasal insertion is a process that is regulated by the presence (or absence) a lexical terminal syntactic node being located in the same phase as the DP that is spelled-out. In this way, nasal insertion is a morpho-phonological process that, crucially, makes reference to syntactic contexts. Given that (64) is sensitive to the syntactic context in this way— i.e. whether a lexical head is in the same phase as the nominal, I propose that (64) applies early in the PF module when terminals are replaced with Vocabulary Items. Specifically, when the terminals inside the DP are associated with their corresponding Vocabulary Item(s), if they are spelled-out within a phase with a lexical head, e.g. V, (64) applies, and the DP is realized with a nasal as its initial segment.[18,19] Subsequently, after (64) is applied in the cycle, in the case of nominals with a stem-initial consonant, lenition is triggered before the nasal deletes (as discussed in §2.2.2).

As was argued in §2.1.1, that argument marking is sensitive to phase boundaries was supported by the alternation observed with argument marking between transitive and intransitive subjects. The former are systematically outside the vP phase and never surface marked; the latter are systematically within the vP phase and always surface marked. I propose the reason that (64) is sensitive to phase boundaries is due to the cyclic nature of Spell-out and that (64) makes reference to syntactic contexts. Consider first intransitive constructions, as in (34). Upon the merger of the next higher phase head, i.e. v, Spell-out is triggered on the DP and terminals within the DP are associated with Vocabulary Items. Subsequently, when the phase head C merges, the next Spell-out cycle occurs, and the vP phase is sent to the interfaces. Since, there is a lexical head V and a DP in the vP phase, (64) is triggered, as illustrated below in (65); Spell-out cycles are boldfaced).

(65) a.	[$_{vP}$ v+V **DP**]	Spell-out of DP phase
b.	[$_{vP}$ v+V [**$_{DP}$** /α/]]	Vocabulary Insertion for DP phase
c.	[$_{CP}$ C **[$_{vP}$ v+V** [$_{DP}$/α/]] …]	Spell-out vP phase
d.	[$_{CP}$ C **[$_{vP}$ v+V** *n*-[$_{DP}$/α/]] …]	*n*-insertion (64)
e.	[$_{CP}$ C **[$_{vP}$** /β/ *n*-[$_{DP}$/α/]] …]	Vocabulary Insertion for vP phase

Consider now the transitive construction, as in (33). When Spell-out is triggered for the vP phase, the lexical V triggers nasal insertion on the object nominal. Moreover, V is also replaced its associated Vocabulary Item. Later, in the next Spell-out cycle, the subject DP is spelled out in the CP phase. However, although the subject DP is spelled-out inside the CP phase, there is no lexical syntactic head that could trigger (64). As (66) shows, when the subject DP is spelled-out, the lexical V has already been replaced with phonological material (which occurred in the earlier

[18] As suggested by an anonymous reviewer, it is worth noting that there is research that has argued against positing a cycle for phrasal phonological processes, e.g. see Cheng & Downing 2016 (see also Selkirk 2011). Subsequently, such approaches reject a direct link between syntactic (i.e. phases) and PF domains. However, there is a body of work that has argued, for a range of morpho-phonological phenomena, that phases play a crucial explanatory role, e.g. Pak 2008; d'Alessandro & Scheer 2015; Bošković 2017; 2025; Fenger 2020; Fenger et al. 2025; Fenger & Weisser 2025; Sande et al. 2020; note, that Harðarson 2022 shows that the data discussed in Cheng & Downing 2016 can, in fact, be analyzed in phasal terms. In this respect, the present analysis is situated within this latter body of work, which hypothesizes that phases play a crucial explanatory role in delimiting when certain morpho-phonological processes may apply.

[19] In this respect, the morpho-phonological processes that underlie mutation in Nias parallel the processes that underlie mutation in Celtic as well. As Pyatt (1997) shows, the morpho-phonological processes underlying mutation in Celtic are sensitive to morpho-syntactic features, and, thus, mutation is argued to occur 'early' in the PF module as well, where such features are present in the derivation.

vP Spell-out cycle). Thus, due to the cyclic nature of Spell-out and that (64) makes reference to syntactic contexts, transitive subjects are systematically unmarked.

(66) a. $[_{CP}$ C $[_{vP}$ /β/ *n*-$[_{DP}$/α/]] … **DP** …] Spell-out of subject DP phase
b. $[_{CP}$ C $[_{vP}$ /β/ *n*-$[_{DP}$/α/]] … $[_{DP}$/γ/] …] Vocabulary Insertion for Subject DP phase

In this respect, Nias' nasal insertion process and its sensitivity to phase boundaries parallels a number of other morpho-phonological process, which exhibit a similar sensitivity. This type of phase domain sensitivity can be observed with several external sandhi processes. As shown in (67a,67b) in Mende the initial segment of the verb undergoes voicing in the presence of an immediately preceding overt object. However, when the object is null (68a) or moves to a vP-external position (68b), this process is neutralized (Cowper & Rice 1987).

(67) a. ndòpóì̀ [$_{VP}$ mbòmèí̀ **v**èmbɛ́ngà]
child hammock swing
'The child swung the hammock.'
b. ndòpóì̀ [$_{VP}$ ngúlɛ́í̀ **g**bàndì̀á]
child oil heated
'The child heated the oil.'

(68) a. ndòpóì̀ [$_{VP}$ *e* **f**èmbɛ́ngà]
child swing
'The child swung it.'
b. gbɛ́mì̀á$_i$ ndòpóì̀ [$_{VP}$ t$_i$ **k**bàndì̀á]
what child heating
'What has the child heated?' (Cowper & Rice 1987: 189,190)

Cowper & Rice (1987) argues that the the key difference between (67a,67b) and (68a,68b) is that while in (67a,67b) the adjacent nominal, i.e. the sandhi trigger, is in the vP/VP with the verb, in (68a,68b) the nominal is vP/VP-external. Assuming that vP delineates a phase, then the Mende data indicate that a requirement on mutation is that the trigger and target be located in the same phase (for similar cases of where sandhi processes are conditioned by phases see: e.g., Pak 2008; d'Alessandro & Scheer 2015, and Bošković 2017).

The same effect occurs in Welsh, where phase boundaries can neutralize mutation. As shown in (69a), the first element after the subject surfaces mutated. However, when the subject and the adjacent element are in distinct CPs, as in (69b) with the *wh*-item, mutation is neutralized (Pyatt 2003).

(69) a. Gwelodd Gwen **g**ath
saw Gwen cat
'Gwen saw a cat' **c**ath → **g**ath
b. Gofynnodd Gwen [$_{CP}$ **p**wy (***b**wy) a welodd gath]
asked Gwen who AFF saw cat
Gwen asked who saw a cat. **p**wy ↛ **b**wy (Pyatt 2003: 214, 215)

Assuming that CP delimits a phase in (69b), what the above data show is that Welsh mutation can be neutralized in cases where the trigger and target occupy distinct phases (see Pyatt 2003 for a proposal of this pattern in terms of intonational phrase boundaries, which have been argued to correspond to phases, cf. Kratzer & Selkirk 2007; Selkirk 2011).

In this respect, I argue that Nias' argument marking pattern is an instance of this more general pattern, where a (morpho)-phonological process can be neutralized in contexts where the trigger

and target are in distinct phases. For Mende, this is observed when the object and verb are both realized in vP. In Welsh, mutation is neutralized when the trigger and target are in distinct phases. For Nias this is observed with transitive subjects, which are outside the vP and, thus, argument marking is neutralized.

3 Conclusion

In this paper, I have demonstrated that Nias' argument marking system is not the realization of a 'marked absolutive' case system but rather is a morpho-phonological process, unrelated to case, that is regulated by independent syntactic considerations in Nias. In particular, that argument marking in Nias displays a domain sensitivity (i.e. phases) that, in certain contexts, results in a superficial 'marked absolutive' pattern. The domain sensitivity of the Nias argument marking process, places the phenomenon in the same class as other morpho-phonological processes, e.g. mutation in Celtic and Mende, that are conditioned whether the trigger and target occupy the same syntactic domain.

That Nias' 'marked absolutive' pattern emerges from a morpho-phonological process unrelated to case is consequential from a typological standpoint. Namely, that since this pattern is simply due to morpho-phonological processes unrelated to case, the only potentially attested instance of a 'marked absolutive' case pattern is, in fact, an illusory one. In this way, the typology of attested case patterns is further restricted. Furthermore, I suggest that Nias is an instructive case study when theorizing about what appears to be a typologically idiosyncratic case pattern. In Nias, what initially looked like an idiosyncratic case pattern—i.e. 'marked absolutive', was revealed to be a conspiracy of otherwise common grammatical processes. Given that such a situation can arise in Nias, this suggests that other idiosyncratic case patterns may be a surface effect due to a confluence of regular processes as well. Thus, I suggest that this study on Nias' 'case pattern', invites future research on, and a reexamination of other idiosyncratic case patterns.

Abbreviations

ASS= associative; COLL= collective; DIST= distal; DYN= dynamic verb-forming prefix; HAVE= verb-forming prefix; MUT= mutated form; RES= resultative verb-forming affix; RLS=realis; ST= stative prefix; TR=transitive verb-forming affix.

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